

Chapter 3

Tuesday, January 30, 2024 8:33 AM

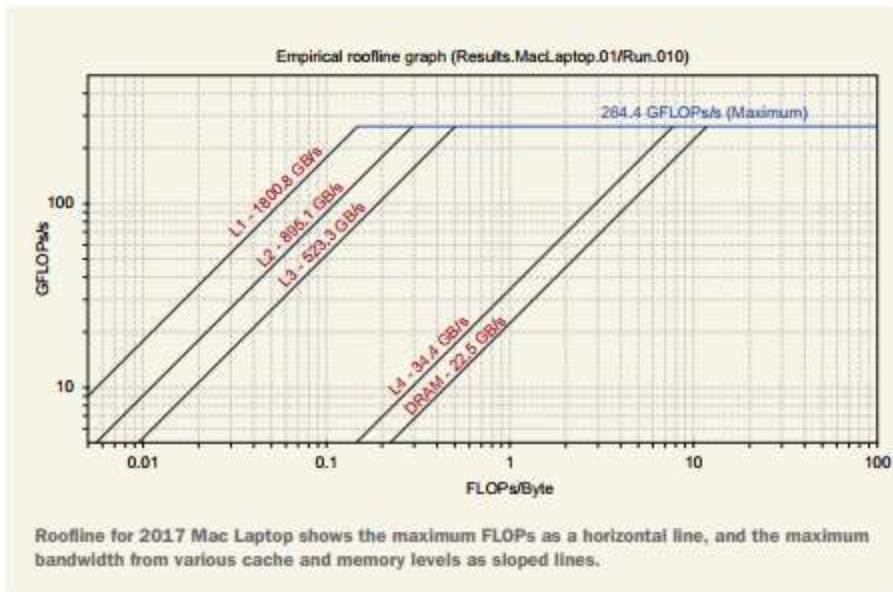
- Hardware can be limiting in multiple ways
 - FLOPS
 - OPS
 - Memory Bandwidth
 - Memory Latency
 - Instruction queue
 - Network
 - Disk
- Speed vs. Feed
 - Speed is how fast ops can be done
 - Feed is how fast information can be provided
- Theoretical limits hard to hit
 - Example: Depending on cache usage, the bandwidth for memory can change
- Benchmarking: FLOPS, GB/s, Watts
- Tools for Processor Information
 - Intel: ark.intel.com
 - AMD: amd.com/en/products/specifications/processors
 - Hwloc gives access to lstopo
 - Lscpu for our cluster:

```
Architecture:      x86_64
CPU op-mode(s):    32-bit, 64-bit
Address sizes:      48 bits physical, 48 bits virtual
Byte Order:         Little Endian
CPU(s):             32
On-line CPU(s) list: 0-31
Vendor ID:          AuthenticAMD
Model name:         AMD EPYC 7551P 32-Core Processor
CPU family:         23
Model:              1
Thread(s) per core: 1
Core(s) per socket: 32
Socket(s):          1
Stepping:           2
Frequency boost:    enabled
CPU max MHz:        2000.0000
CPU min MHz:        1200.0000
BogoMIPS:           3999.77
Flags:              fpu vme de pse tsc msr pae mce cx8 apic sep mtrr pge mca cmov pat pse36 clflush mmx fxsr sse sse2 ht
syscall nx mmxext fxsr_opt pdpe1gb rdtscp lm constant_tsc rep_good nopl nonstop_tsc cpuid extd_apicid amd_dcm
aperfmpperf rapl pni pclmulqdq monitor ssse3 fma
                    cx16 sse4_1 sse4_2 movbe popcnt aes xsave avx f16c rdrand lahf_lm cmp_legacy svm extapic cr8_legacy
abm sse4a misalignsse 3dnowprefetch osvw skinit wdt tce topoext perfctr_core perfctr_nb bpext perfctr_llc mwaitx cpb
hw_pstate ssbd ibpb vmcall fsgsbase bmi1
                    avx2 smep bmi2 rdseed adx smap clflushopt sha_ni xsaveopt xsavec xgetbv1 clzero irperf xsaveerptr arat
npt lbrv svm_lock nrip_save tsc_scale vmcb_clean flushbyasid decodeassists pausefilter pfthreshold avic
v_vmsave_vmload vgif overflow_recov succor smca
Virtualization features:
Virtualization:     AMD-V
Caches (sum of all):
L1d:                1 MiB (32 instances)
L1i:                2 MiB (32 instances)
L2:                 16 MiB (32 instances)
L3:                 64 MiB (8 instances)
```

NUMA:

NUMA node(s): 4
NUMA node0 CPU(s): 0-7
NUMA node1 CPU(s): 8-15
NUMA node2 CPU(s): 16-23
NUMA node3 CPU(s): 24-31

- FLOPS maximum can be calculated as $\text{Cores} * \text{Clock Rate} * \text{Flops/Cycle}$
 - If multiple operations can be performed per cycle, need to add a factor for that
 - Example: Fused multiply add
- Memory Bandwidth maximum calculated as $\text{Memory transfer rate} * \text{channels} * \text{bytes} * \text{socket}$
 - Better to measure empirically
- STREAM tool can be used for empirical measurement
 - <https://github.com/jeffhammond/STREAM.git>
 - This repo belongs to Jeff Hammond, know him via CSGF
- Roofline plot can show how memory limits FLOPS



- Profiling tools allow us to measure a working program
 - Call graphs allow us to know hot spots in the program
 - Cachegrind
 - KCacheGrind / QCacheGrind show results
 - KCacheGrind uses X11
 - Intel advisor is a commercial tool, but is now part of OneAPI which we have installed.
- Chapter contains information on plotting your own Roofline plots, not copying in detail here
 - Uses matplotlib / python
 - <https://github.com/EssentialsofParallelComputing/Chapter3>
- Tracking memory usage is tricky
 - MemSTATS calls can be put in the program directly